



Problem Statement:

Given a sorted array and a value x, the floor of x is the largest element in array smaller than or equal to x. Write divide and conquer algorithm to find floor of x.

Input Format

First Line Contains Integer n – Size of array

Next n lines Contains n numbers – Elements of an array

Last Line Contains Integer x – Value for x

Output Format

First Line Contains Integer – Floor value for x

Answer: (penalty regime: 0 %)

```

1  #include <stdio.h>
2
3  int findFloor(int arr[], int low, int high, int x) {
4      if (low > high) {
5          return -1;
6      }
7
8      if (x >= arr[high]) {
9          return arr[high];
10     }
11
12     int mid = low + (high - low) / 2;
13
14     if (arr[mid] == x) {
15         return arr[mid];
16     }
17
18     if (mid > 0 && arr[mid - 1] <= x && x < arr[mid]) {
19         return arr[mid - 1];
20     }
21
22     if (x < arr[mid]) {
23         return findFloor(arr, low, mid - 1, x);
24     }
25
26     return findFloor(arr, mid + 1, high, x);
27 }
28
29 int main() {
30     int n;
31     scanf("%d", &n);
32     int arr[n];
33     for (int i = 0; i < n; i++) {
34         scanf("%d", &arr[i]);
35     }
36
37     int x;
38     scanf("%d", &x);
39
40     int floorVal = findFloor(arr, 0, n - 1, x);
41     printf("%d\n", floorVal);
42
43     return 0;
44 }
45

```

Check

	Input	Expected	Got	
✓	6 1 2 8 10 12 19 5	2	2	✓
✓	5 10 22 85 108 129 100	85	85	✓
✓	7 3 5 7 9 11 13 15 10	9	9	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

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